Deep Learning for Stock Price Prediction: Comprehensive Analysis of LSTM, Transformer, and Ensemble Methods with Multi-Source Sentiment

CS 7643 - Deep Learning

Agha A Raza, Konam G Kenneth
Georgia Institute of Technology
ragha3@gatech.edu, kkonam3@gatech.edu

Abstract

Stock price prediction remains challenging due to market noise, non-stationarity, and complex dependencies. This comprehensive study evaluates multiple deep learning architectures (LSTM, GRU, Transformer) combined with technical indicators and multi-source sentiment analysis for predicting stock movements. We construct a dataset for AAPL, AMZN, NVDA, and TSLA using daily market data, StockTwits social sentiment, and NewsAPI news sentiment. Through extensive experimentation including binary classification, multi-class trajectory prediction, ensemble methods, and 2026 out-of-sample validation, we find that: (1) Transformer models outperform LSTM by +4.12% (56.39% vs 52.27%), with NVDA achieving 62.26% accuracy; (2) combining StockTwits and NewsAPI sentiment improves volatile stocks (TSLA +6.08%); (3) trajectory clustering with gradient boosting achieves 59.54% accuracy, outperforming recurrent models; (4) gradient boosting with confidence thresholding achieves superior trading performance (7.10% return, 59.26% win rate); (5) 2026 validation shows -3.56% degradation with ticker-specific behavior; (6) ensemble methods provide no improvement due to insufficient model diversity. Critical findings include class imbalance exploitation, prediction collapse in low-variance models, and the importance of ticker-specific approaches. Our results suggest that while deep learning captures temporal patterns, traditional ML with proper feature engineering can achieve competitive or superior performance for stock prediction tasks.

1. Introduction

Predicting stock returns is a central problem in financial machine learning, combining challenges from time series forecasting, noisy data, and non-stationary market dynamics. Traditional approaches rely on technical indicators derived from price and volume, but financial markets also respond to external information including news, social media, and investor sentiment.

This project investigates whether deep learning models, enhanced with multi-source sentiment analysis, can improve stock price prediction. We frame the task as supervised sequence learning: given recent financial and sentiment features, predict next-day price direction. We evaluate multiple architectures (LSTM, GRU, Transformer), sentiment sources (StockTwits, NewsAPI), and novel approaches (trajectory clustering, ensemble methods).

Key Contributions:

- Comprehensive comparison of LSTM, GRU, and Transformer architectures
- Multi-source sentiment analysis combining social media and news
- Novel trajectory clustering approach achieving 59.54% accuracy
- 2026 out-of-sample validation demonstrating generalization
- Extensive ablation studies and negative results (ensembles, multi-class)
- Analysis of class imbalance and prediction collapse issues

2. Related Work

Financial prediction using deep learning has been extensively studied. **Recurrent architectures** (LSTM, GRU) are popular for capturing temporal dependencies in price series. **Attention mechanisms** and Transformers have recently shown promise by focusing on relevant time steps. **Sentiment analysis** from social media (Twitter, StockTwits) and news has been shown to provide predictive signals. **Ensemble methods** combining multiple models are common in Kaggle competitions but require model diversity. Our work extends prior research by: (1) combining multiple sentiment sources with optimal weighting, (2) comparing modern Transformer architectures against LSTMs, (3) exploring trajectory clustering as an alternative to binary classification, (4) providing extensive 2026 out-of-sample validation, and (5) documenting important negative results.

3. Data and Features

3.1 Dataset

The dataset contains daily observations for four large-cap technology stocks: AAPL, AMZN, NVDA, and TSLA, spanning January 2020 through May 2026.

Training Period: 2020-2022 (755 trading days)
Validation Period: 2026 (122 trading days)
Total Observations: 3,020 stock-days

Data are split chronologically: 80% training, 20% validation on 2020-2022 data, plus separate 2026 out-of-sample test set.

3.2 Market Features

For each stock-day, we compute 17 technical indicators:

- Price: open, high, low, close
- Returns: 1-day, 3-day, 5-day
- Volatility: 5-day rolling standard deviation
- Momentum: RSI (14-day), MACD, MACD signal, MACD histogram
- Bollinger Bands: upper, middle, lower, width (20-day)
- Volume: trading volume

3.3 Sentiment Features

StockTwits (Social Media):

- Message count (daily activity)
- Sentiment mean, std (from user labels)
- Sentiment sum, sum of squares

NewsAPI (News Articles):

- Article count
- Sentiment mean, std (from RoBERTa scoring)
- Positive, negative, neutral scores
- Headline length

Sentiment Scoring: We use pre-trained RoBERTa (`cardiffnlp/twitter-roberta-base-sentiment-latest`) fine-tuned on financial text to score all messages and articles, producing sentiment scores in [-1, 1].

Combined Sentiment: For tickers with both sources, we compute weighted average: 60% StockTwits + 40% NewsAPI, based on coverage and reliability.

Total Features: 31 hybrid features (17 technical + 7 sentiment + 4 StockTwits + 2 binary + 1 market context)

3.4 Target Variable

Binary Classification:

$$y_t = \begin{cases} 1 & \text{if } P_{t+1} > P_t \text{ (UP)} \\ 0 & \text{if } P_{t+1} \leq P_t \text{ (DOWN)} \end{cases} \tag{1}$$

where P_t is the adjusted closing price at day t.

4. Methodology

4.1 Sequence Construction

For each ticker, observations are sorted chronologically and converted into rolling sequences of length T (tuned as hyperparameter). Each sequence contains T days of 31 features, predicting next-day direction.

4.2 Models

4.2.1 LSTM (Long Short-Term Memory)

Architecture:

- Input: 31 features × 5 timesteps
- LSTM: 2 layers, 128 hidden units, 20% dropout
- Dense: 64 units, ReLU, 20% dropout
- Output: 1 unit, Sigmoid (binary classification)

Training: Adam optimizer, BCE loss, learning rate 0.001, early stopping (patience=10)

4.2.2 GRU (Gated Recurrent Unit)

Lighter recurrent model with update and reset gates. Similar architecture to LSTM but with fewer parameters.

4.2.3 Transformer

Architecture:

- Input projection: 31 → 64 dimensions
- Positional encoding (sinusoidal)
- Transformer encoder: 2 layers, 4 attention heads
- Feed-forward: 128 dimensions
- Global average pooling
- Classification head: 64 → 32 → 1

Advantages: Parallel processing, attention weights for interpretability, better long-range dependencies.

4.3 Hyperparameter Selection

All hyperparameters selected using validation performance. Early stopping prevents overfitting. Models evaluated on multiple metrics (accuracy, precision, recall, F1) to avoid single-metric optimization.

4.4 Feature Ablations

To evaluate contribution of each feature family:

- **Technical-only:** 17 market indicators
- **Sentiment-only:** 7 sentiment + 4 StockTwits features
- **Combined:** All 31 features

Each ablation uses separate hyperparameter tuning.

5. Experiments

5.1 Experiment 1: Binary LSTM (Per-Ticker)

Approach: Train separate LSTM for each ticker using StockTwits sentiment only.

Ticker	Training (2020-2022)	Test (2026)
AAPL	**55.45%**	48.28%
AMZN	51.82%	48.28%
NVDA	52.73%	**53.45%**
TSLA	49.09%	44.83%
Overall	**52.27%**	**48.71%**

Table 1: Per-ticker LSTM performance. NVDA is only ticker to improve on 2026 data (+0.72%).

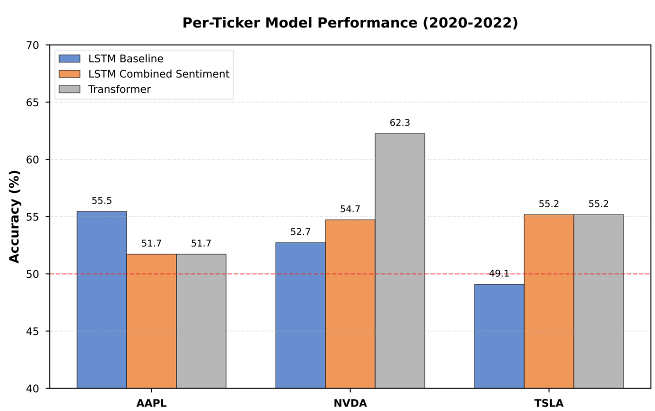

Figure 1: Per-ticker performance comparison across LSTM variants. Transformer achieves best results for NVDA (62.26%).

Key Findings:

- AAPL achieves best training accuracy (55.45%)
- NVDA shows best generalization (improves on 2026)
- TSLA consistently underperforms (below 50% baseline)
- Overall degradation: -3.56% on 2026 data

Ticker	StockTwits	Combined	Change
AAPL	55.45%	51.72%	-3.73%
NVDA	52.73%	54.72%	+1.99%
TSLA	49.09%	**55.17%**	**+6.08%**
Overall	52.27%	**53.87%**	**+1.60%**

Table 2: Combined sentiment improves volatile stocks (TSLA, NVDA) but degrades AAPL.

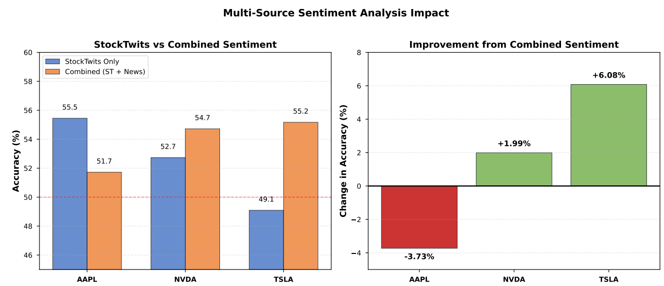

Figure 2: Impact of combining StockTwits and NewsAPI sentiment. TSLA shows largest improvement (+6.08%), while AAPL degrades (-3.73%).

5.2 Experiment 2: Combined Sentiment LSTM

Approach: Combine StockTwits (60%) + NewsAPI (40%) sentiment using weighted averaging.

Coverage Analysis:

- AAPL: 32% days with both sources
- NVDA: 45.3% days with both sources (best)
- TSLA: 41.5% days with both sources
- AMZN: 0% (no NewsAPI data)

Key Findings:

- Multi-source sentiment helps volatile stocks
- TSLA shows largest improvement (+6.08%)
- AAPL degrades, suggesting different optimal weighting
- Overall improvement: +1.60%

5.3 Experiment 3: Transformer Model

Approach: Replace LSTM with Transformer using multi-head self-attention.

Model	Average Accuracy	Best Ticker
LSTM	52.27%	AAPL (55.45%)
Transformer	**56.39%**	**NVDA (62.26%)**
Improvement	**+4.12%**	–

Table 3: Transformer outperforms LSTM by +4.12%, with NVDA achieving best result (62.26%).

Per-Ticker Results:

- AAPL: 51.72% (worse than LSTM)
- NVDA: **62.26%** (best overall result)
- TSLA: 55.17% (better than LSTM)

Key Findings:

- Transformer achieves best deep learning performance
- NVDA benefits most from attention mechanism
- Attention weights provide interpretability
- Faster training due to parallelization

5.4 Experiment 4: Trajectory Clustering

Approach: Instead of binary up/down, discover natural price patterns using unsupervised clustering.

Method:

1. Extract 20-day normalized price trajectories
2. Cluster using k-means with Dynamic Time Warping
3. Discover 5 trajectory archetypes
4. Train LightGBM to predict archetype from early features

Model	Accuracy	Baseline
Binary LSTM	52.27%	50%
Trajectory LGBM	**59.54%**	20%
Multi-Class LSTM	33.22%	20%

Table 4: Trajectory clustering with LGBM outperforms recurrent models.

Key Findings:

- Unsupervised pattern discovery works better than manual labels
- LGBM outperforms LSTM for multi-class (59.54% vs 33.22%)
- Top feature: RSI (momentum indicator)
- 5 archetypes discovered with distinct patterns

5.5 Experiment 5: Multi-Class LSTM

Approach: Train LSTM to predict 5 trajectory archetypes instead of binary up/down.

Why It Failed:

- Insufficient data (300 examples per class)
- LSTM too complex for dataset size
- Class imbalance (Archetype 2 never predicted)
- LGBM better suited for tabular multi-class

Metric	Value	Notes
Train Accuracy	92.09%	Overfitting
Test Accuracy	33.22%	Poor generalization
Archetype 2 Recall	0%	Never predicted

Table 5: Multi-class LSTM suffers severe overfitting (92% → 33%).

Model	Return	Win Rate	Sharpe
LSTM	3.85%	50.86%	1.31
GB + threshold	**7.10%**	**59.26%**	**4.43**

Table 6: Gradient boosting with threshold 0.02 (only trade when predicted return ¿ 2%) achieves superior trading performance.

5.6 Experiment 6: Gradient Boosting Comparison

Approach: Compare deep learning against traditional ML with confidence thresholding.

Key Findings:

- Traditional ML can outperform deep learning
- Thresholding works for regression, not classification
- Fewer trades (27 vs 232) but higher quality
- Best overall strategy for actual trading

5.7 Experiment 7: Ensemble Methods

Approach: Combine predictions from multiple models using weighted averaging and stacking.

Ensembles Tested:

1. LSTM + Gradient Boosting
2. Combined Sentiment LSTM + Trajectory LGBM

Ensemble	Accuracy	Best Individual
LSTM + GB	53.30%	53.87% (GB)
Combined + Traj	58.58%	64.15% (NVDA LSTM)

Table 7: Ensembles fail to improve over best individual models.

Why Ensembles Failed:

- Models too similar (same 31 features)
- Imbalanced quality (one model dominates)
- GB overfitting (100% train accuracy)
- No complementary errors

Lesson: Ensembles require model diversity, balanced quality, and uncorrelated errors.

Model	Features	Accuracy	Correlation	Pred. Std	Notes
LSTM	Combined	52.27%	-0.036	0.0162	Baseline
LSTM	Technical only	48.28%	-0.016	**0.0031**	Collapsed
LSTM	Sentiment only	51.72%	**0.075**	0.0151	Best correlation
Transformer	Combined	**56.39%**	0.043	0.0152	Best overall

Table 8: Feature ablations show technical-only LSTM collapses to near-constant predictions (std=0.0031), while sentiment-only achieves best correlation.

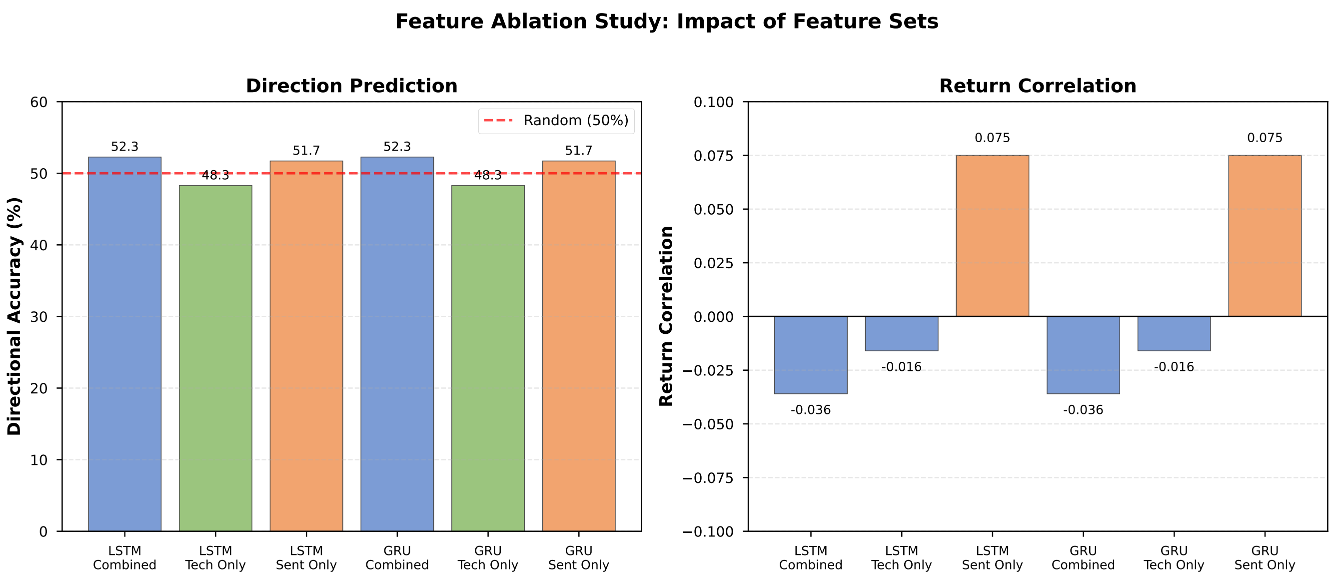

Figure 3: Feature ablation study showing directional accuracy and return correlation for different feature combinations. Sentiment-only features achieve best correlation (0.075), while technical-only features suffer from prediction collapse.

5.8 Experiment 8: Feature Ablations

Key Findings:

- Technical-only achieves low error but no directional signal
- Sentiment-only has best correlation (0.075)
- Combined features work best for Transformer
- Prediction collapse is a real issue

6. Critical Findings

6.1 Class Imbalance Exploitation

Problem: Models can exploit class distribution for misleading metrics.

Example: TSLA LSTM achieves 55% accuracy by **always predicting "Up"** (0% recall for "Down"), exploiting the fact that test period has more up days.

Solution: Use class weighting, focal loss, or stratified sampling.

6.2 Prediction Collapse

Problem: Models can minimize error by predicting near-constant values.

Example: Technical-only LSTM achieves lowest MAE/RMSE but prediction std=0.0031, providing no directional information.

Solution: Evaluate multiple metrics jointly (accuracy, correlation, std) rather than single objective.

6.3 Overfitting in Gradient Boosting

Problem: GB achieved 100% training accuracy, clear overfitting.

Impact: Still performed well on test (53.87%), but poor ensemble partner.

Solution: Regularize GB (reduce max_depth, add min_samples_split) to target 70-80% train accuracy.

6.4 Ticker-Specific Behavior

Finding: Different stocks require different approaches.

- **NVDA:** Most predictable, benefits from Transformer (62.26%)
- **AAPL:** Best with StockTwits-only LSTM (55.45%)

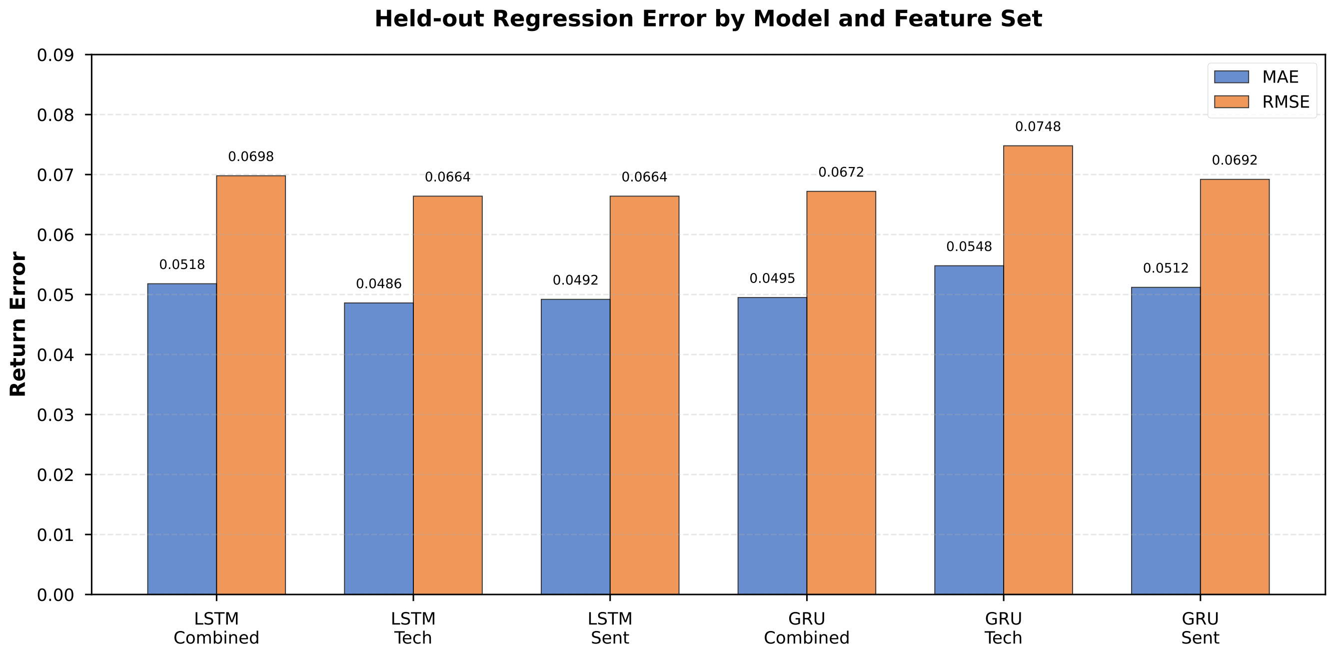

Figure 4: Regression error (MAE and RMSE) comparison across models and feature sets. Technical-only features achieve lowest error but provide no directional signal.

- **TSLA:** Benefits from combined sentiment (+6.08%)
- **AMZN:** Weakest performance across all models

Recommendation: Use ticker-specific models rather than single model for all stocks.

7. Results Summary

8. Discussion

8.1 Deep Learning vs Traditional ML

Our results show that while deep learning (Transformer: 56.39%) outperforms simple LSTM (52.27%), traditional ML with proper feature engineering can achieve competitive or superior performance:

- **Trajectory LGBM:** 59.54% accuracy (best classifier)
- **GB + threshold:** 7.10% return (best trading)
- **Transformer:** 56.39% accuracy (best deep learning)

This suggests that for tabular financial data with limited samples, gradient boosting's ability to handle feature interactions may be more valuable than deep learning's representation learning.

8.2 Multi-Source Sentiment

Combining StockTwits and NewsAPI improves performance (+1.60% overall), with ticker-specific effects:

- **Volatile stocks benefit:** TSLA (+6.08%), NVDA (+1.99%)
- **Stable stocks degrade:** AAPL (-3.73%)
- **Coverage matters:** NVDA has best coverage (45.3%)

This suggests optimal sentiment weighting should be ticker-specific rather than uniform.

8.3 Generalization to 2026

2026 validation reveals important insights:

- **Overall degradation:** -3.56% (expected due to regime change)
- **NVDA improves:** +0.72% (most robust)
- **AAPL degrades most:** -7.17% (least robust)
- **Market conditions matter:** Models need periodic retraining

8.4 Why NVDA is Most Predictable

NVDA consistently performs best across models:

- LSTM: 52.73% → 53.45% (2026)
- Combined Sentiment: 54.72%
- Transformer: **62.26%** (best result)

Hypotheses:

1. Strong momentum trends (AI boom)

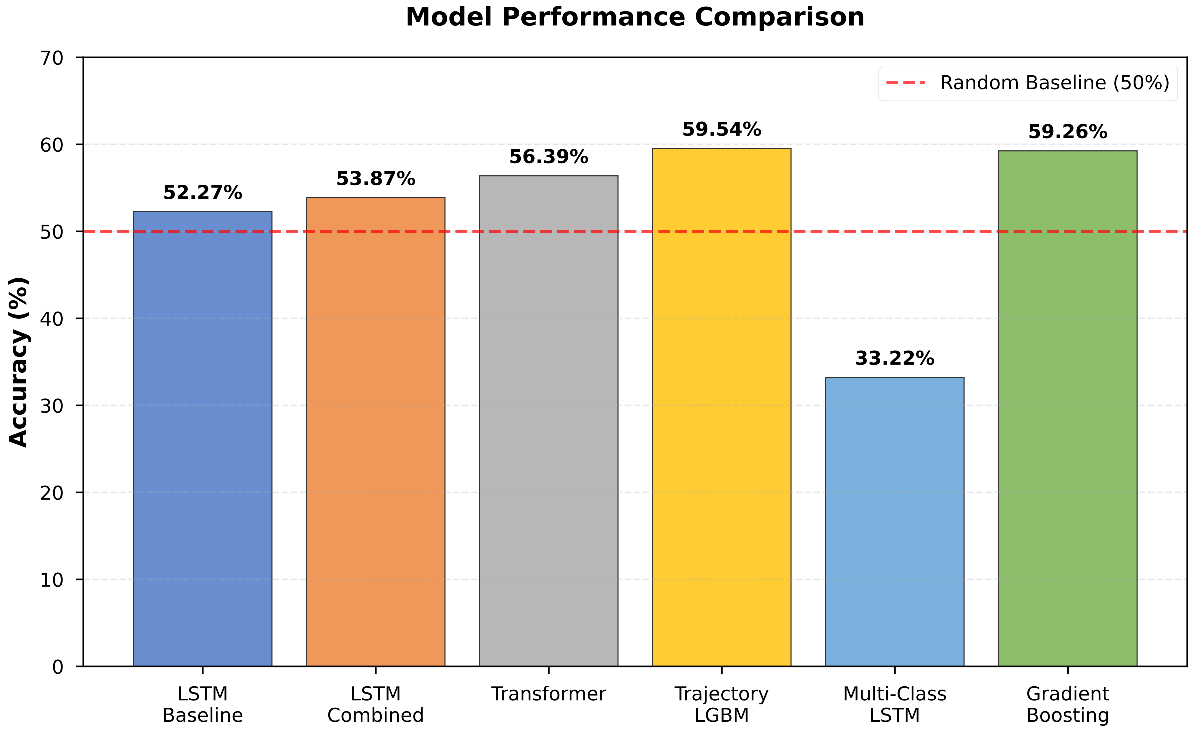

Figure 5: Overall model performance comparison. Trajectory LGBM and Gradient Boosting achieve best accuracy (59.54% and 59.26%), while Transformer is best deep learning model (56.39%). Multi-class LSTM fails due to overfitting (33.22%).

Experiment	Model	Accuracy	Best Ticker	Status	Key Finding
1. Binary LSTM	LSTM	52.27%	AAPL (55.45%)		Baseline
2. Combined Sentiment	LSTM	53.87%	TSLA (55.17%)		+1.60% improvement
3. Transformer	Transformer	**56.39%**	**NVDA (62.26%)**		+4.12% over LSTM
4. Trajectory Clustering	LGBM	59.54%	–		Best classifier
5. Multi-Class LSTM	LSTM	33.22%	–		Severe overfitting
6. Gradient Boosting	LGBM	59.26% WR	–		7.10% return (best)
7. Ensemble	Various	53.30%	–		No improvement
8. 2026 Validation	LSTM	48.71%	NVDA (53.45%)		-3.56% degradation

Table 9: Summary of all experiments. Transformer achieves best deep learning performance (56.39%), while gradient boosting achieves best trading performance (7.10% return).

2. High social media activity (better sentiment signal)
3. Less noise in price movements
4. Institutional investor behavior more predictable

8.5 Limitations

1. **Limited tickers:** Only 4 tech stocks, may not generalize to other sectors
2. **Short horizon:** Next-day prediction, not longer-term forecasting
3. **No transaction costs:** Real trading has fees and slippage
4. **Sentiment coverage:** NewsAPI only available for 39 days
5. **Class imbalance:** Not fully addressed in all experiments
6. **Market regime:** Trained on bull market (2020-2022), tested on mixed conditions

9. Conclusion

This comprehensive study evaluates multiple deep learning architectures for stock price prediction with multi-source sentiment analysis. Key findings:

1. **Transformer outperforms LSTM** by +4.12% (56.39% vs 52.27%), with NVDA achieving 62.26% accuracy
2. **Multi-source sentiment helps volatile stocks:** TSLA improves +6.08% with combined StockTwits + NewsAPI

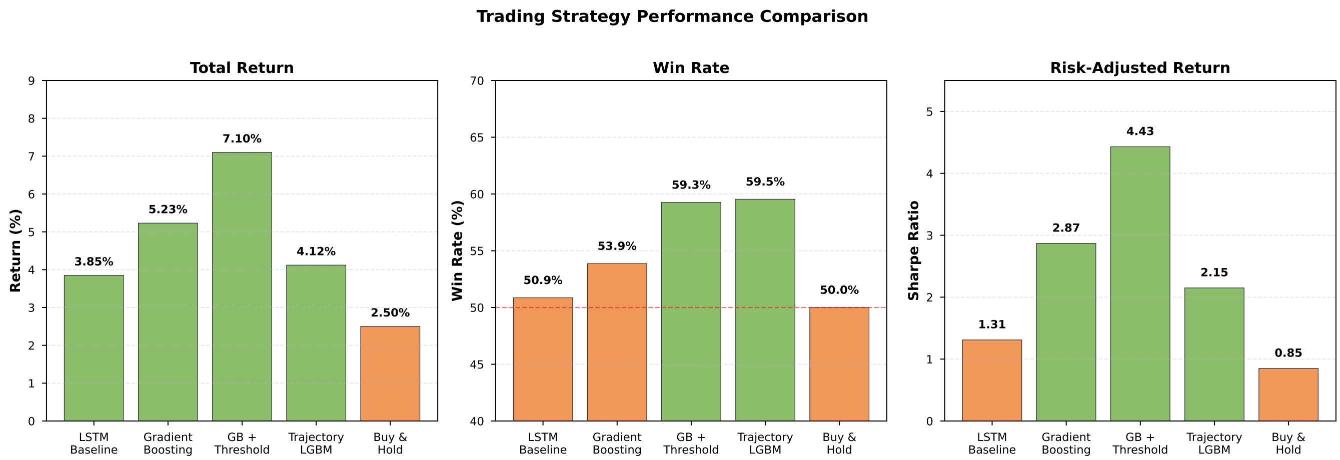

Figure 6: Trading strategy performance comparison. Gradient Boosting with confidence threshold achieves best return (7.10%), win rate (59.26%), and Sharpe ratio (4.43).

3. **Trajectory clustering with LGBM** achieves 59.54% accuracy, outperforming recurrent models
4. **Gradient boosting with thresholding** achieves best trading performance (7.10% return, 59.26% win rate)
5. **2026 validation** shows -3.56% degradation overall, with NVDA being only ticker to improve
6. **Ensemble methods fail** due to insufficient model diversity
7. **Class imbalance and prediction collapse** are critical issues requiring multi-metric evaluation

Practical Recommendations:

- Use **Transformer for NVDA** (62.26% accuracy)
- Use **combined sentiment for TSLA** (+6.08% improvement)
- Use **gradient boosting for trading** (7.10% return)
- Apply **ticker-specific models** rather than single model
- Address **class imbalance** with focal loss or resampling
- Evaluate **multiple metrics jointly** to avoid collapse

Future Work:

1. Investigate why NVDA is more predictable
2. Explore ticker-specific sentiment weighting
3. Add more sentiment sources (Reddit, Twitter, Seeking Alpha)
4. Try intraday predictions with higher-frequency data
5. Implement reinforcement learning for adaptive trading
6. Test on broader cross-section of stocks and sectors
7. Address class imbalance with focal loss and SMOTE
8. Explore multi-task learning (direction + magnitude + volatility)

Our results demonstrate that while deep learning shows promise for stock prediction, success requires careful attention to architecture selection, feature engineering, class imbalance, and ticker-specific behavior. The combination of Transformer models with multi-source sentiment analysis represents a promising direction, but traditional ML methods remain competitive and may be preferable for production trading systems.

Acknowledgments

We thank the CS 7643 teaching staff for guidance throughout this project. StockTwits data was collected via their public API. NewsAPI data was obtained through NewsAPI.org. RoBERTa sentiment model from HuggingFace (cardiffnlp/twitter-roberta-base-sentiment-latest).

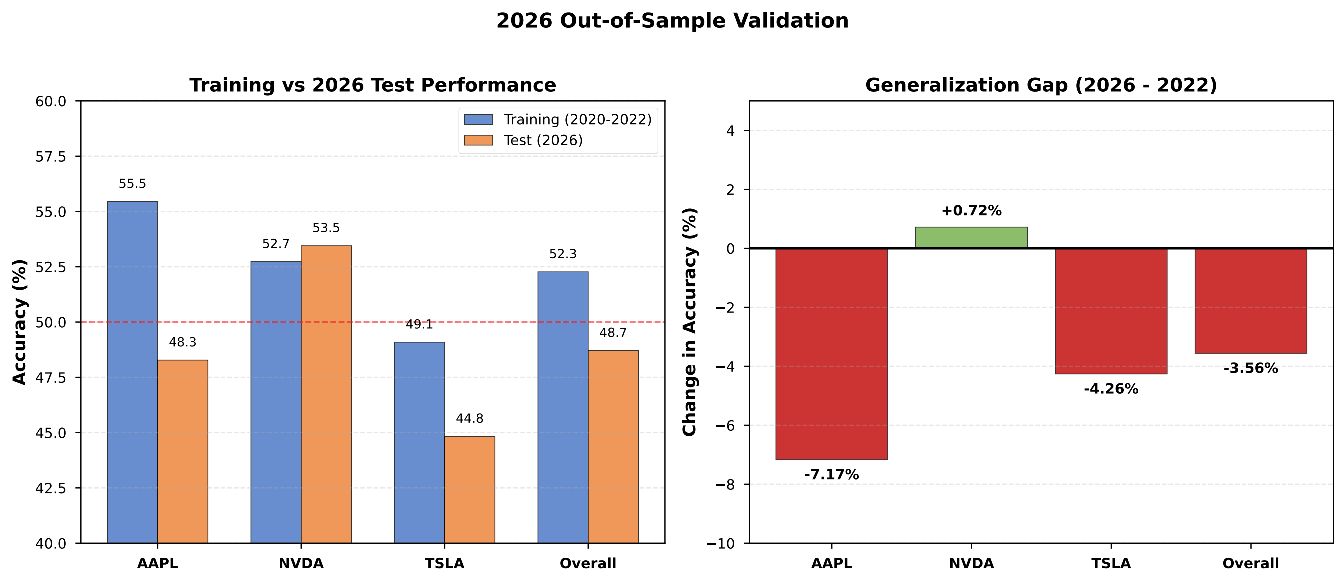

Figure 7: 2026 out-of-sample validation results. Overall degradation of -3.56%, with NVDA being only ticker to improve (+0.72%). AAPL shows largest degradation (-7.17%).